

USE-CASE FLOW SPECIFICATION

UC-03: Allocate ICU Bed

1. Use Case Information

- Use Case ID: UC-03
- Use Case Name: Allocate ICU Bed
- Primary Actor: Triage Nurse
- Supporting Actor: Hospital Administrator
- Goal: Assign an appropriate available ICU bed based on triage severity and current availability.

2. Preconditions

- Patient intake vitals and triage information are recorded.
- A triage severity score has been computed.
- Current ICU bed-status data is available.
- An authorized triage nurse is logged in.

3. Main Success Scenario

1. Triage Nurse selects the patient from the ICU allocation waitlist.
2. The system displays the patient's triage severity and priority.
3. The system checks current ICU bed availability and status.
4. The system identifies a suitable available ICU bed.
5. Triage Nurse confirms the proposed ICU bed allocation.
6. The system marks the selected bed as Occupied and records the allocation.
7. The updated bed status is propagated to connected dashboard clients.
8. The system displays an allocation confirmation.

4. Postconditions

- The patient is assigned to an ICU bed.
- The selected ICU bed is marked Occupied.
- The allocation is recorded for authorized staff to view.
- The dashboard reflects updated ICU availability.

5. Alternate Flow – No Suitable ICU Bed Available

1. At Step 4, the system finds no suitable available ICU bed.
2. The system displays a no-bed-available message.
3. The patient remains on the prioritized ICU allocation waitlist.
4. The system notifies relevant staff that an ICU transfer requires attention.
5. When a suitable bed becomes available, the patient can be reconsidered according to the updated priority order.

6. UML Relationship Context

- UC-03 Allocate ICU Bed «include» UC-01 Compute Triage Severity.
- UC-07 Notify Staff «extend» UC-05 Initiate Ambulance Transfer.